

Chern-Simons Gravity and Neutrino Self-Interactions

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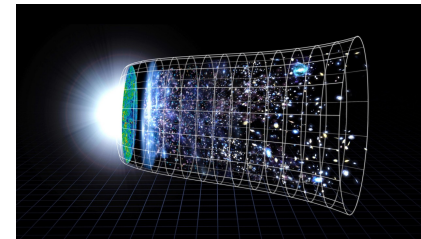
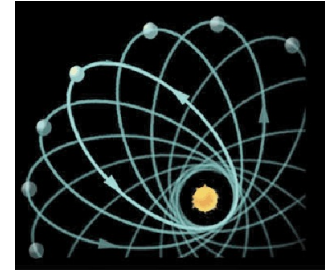
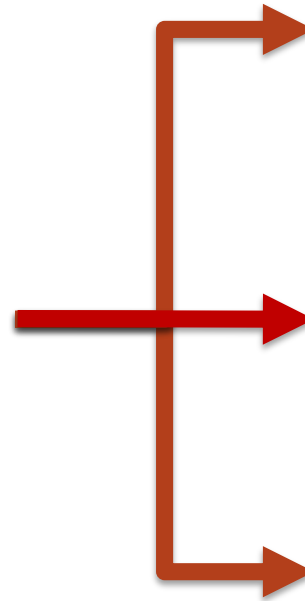
arXiv: 2207.05094, Collaborator: Stephon Alexander

General Relativity: The Einstein-Hilbert Action

$$S = \int d^4x \sqrt{-g} \kappa R$$

Spacetime
Metric,
Lorentzian
Signature

Ricci Scalar,
 $\kappa = (16\pi G)^{-1}$



What is (dynamical) Chern-Simons Gravity [dCS]?

$$S = \int d^4x \sqrt{-g} \left[\kappa R + \frac{a}{4f} {}^*R R - \frac{1}{2} (\nabla_\mu a) (\nabla^\mu a) \right]$$

↑

Spacetime
Metric,
Lorentzian
Signature

↑

Einstein-
Hilbert Term,
 $\kappa = (16\pi G)^{-1}$

↑

dCS Decay
Constant

↑

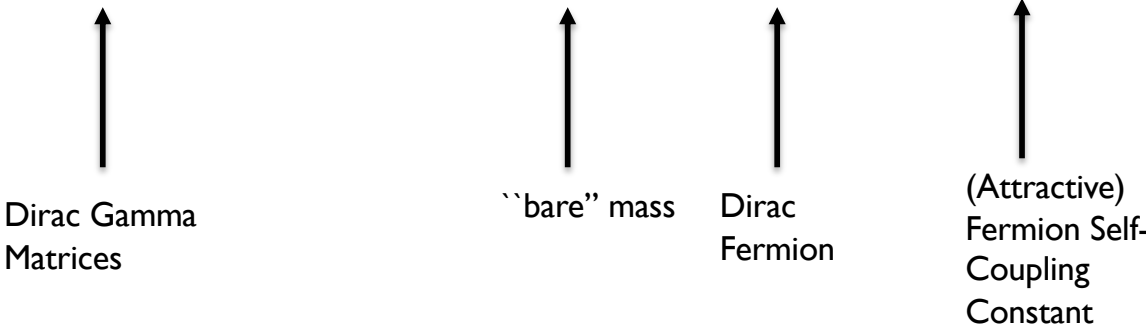
Pontryagin
Density
 ${}^*R R = {}^*R^\rho{}_\sigma{}^{\mu\nu} R^\sigma{}_{\rho\mu\nu}$
 ${}^*R^\rho{}_\sigma{}^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\alpha\beta} R^\rho{}_{\sigma\alpha\beta}$

↑

dCS (pseudo)scalar

What are Fermion Self-Interactions?

$$\mathcal{L}_\Psi = \bar{\Psi} (i\gamma^\mu \partial_\mu - \tilde{m}_\Psi) \Psi - \lambda \bar{\Psi} \Psi \bar{\Psi} \Psi$$



Dirac Gamma Matrices "bare" mass Dirac Fermion (Attractive) Fermion Self-Coupling Constant

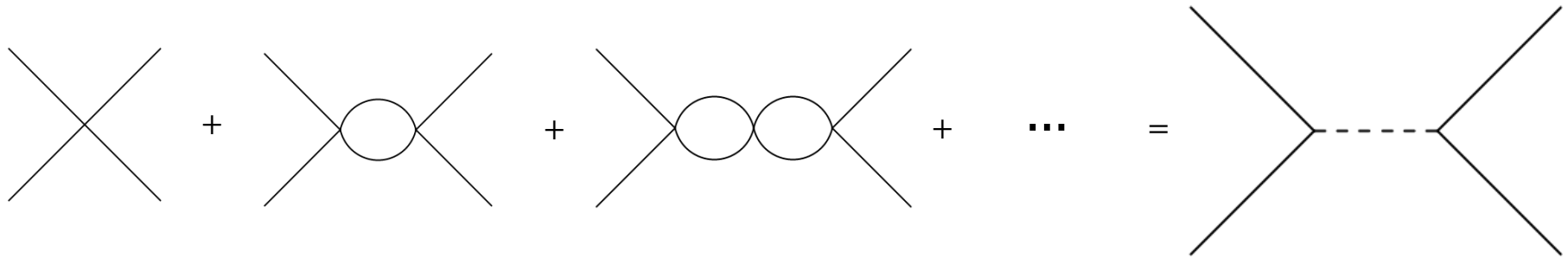
 Cutoff Λ

How will we obtain dCS from Fermion-SI?

1. Identify and realize a scalar bound state in our theory of Fermion Self-Interactions (SI)
2. Demonstrate that the angular part of the (complex) scalar becomes the dCS scalar through fermion-loops with gravity after spontaneous symmetry breaking.

How is there a bound state?

Geometric Summation to identify pole



How to realize the bound state?

$$\begin{aligned}
 Z_\alpha &= \int \mathcal{D}\alpha \mathcal{D}\bar{\alpha} \exp \left(- \int d^4x \tilde{m}_\Phi^2 \bar{\alpha} \alpha \right) \\
 &+ \\
 \Psi &= \Psi_\ell + \Psi_s \\
 \Phi &= \alpha - \tilde{m}_\Phi^{-2} \bar{\Psi}_s \Psi_s \\
 &+
 \end{aligned}
 \Rightarrow$$

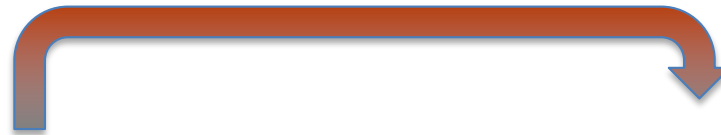
$$\begin{aligned}
 \tilde{\mathcal{L}}_\Psi &= \bar{\Psi} (i\gamma^\mu \partial_\mu - \tilde{m}_\Psi) \Psi - \lambda \bar{\Psi} \Psi \bar{\Psi} \Psi \\
 &+ (\partial_\mu \Phi^*) (\partial^\mu \Phi) - y_\Phi (\Phi \bar{\Psi} \Psi + \text{h.c.}) \\
 &+ m_\Phi^2 |\Phi|^2 - \frac{\lambda_\Phi}{4} |\Phi|^4,
 \end{aligned}$$

Integrate Out Short – Scale Modes

$$y_\Phi \sim \lambda \Lambda^2$$

Spontaneous Symmetry Breaking

$$\Phi = (1/\sqrt{2})(F + \sigma) \exp[ia(x)/F]$$



$$\mathcal{L}_\Phi = \bar{\Psi}_L (i\gamma^\mu \partial_\mu - \tilde{m}_\Psi) \Psi_L + \bar{\Psi}_R (i\gamma^\mu \partial_\mu - \tilde{m}_\Psi) \Psi_R \\ + \partial_\mu \Phi \partial^\mu \Phi^* - V(|\Phi|^2) - (y_\Phi \Phi \bar{\Psi}_L \Psi_R + \text{h.c.}),$$

$$\mathcal{L}_\Phi = \bar{\Psi} \left[i\gamma^\mu \partial_\mu - \left(\tilde{m}_\Psi + \frac{y_\Phi F}{\sqrt{2}} \right) \right] \Psi \\ + \frac{1}{2} \partial_\mu a \partial^\mu a - \frac{y_\Phi}{\sqrt{2}} a \bar{\Psi} \gamma^5 \Psi,$$



The Gap Equation



$$\frac{\Delta m_\Psi}{m_\Psi} \left(\frac{2\pi^2}{\lambda\Lambda^2} \right) = 1 - \frac{m_\Psi^2}{\Lambda^2} \log \left(1 + \frac{\Lambda^2}{m_\Psi^2} \right), \quad \Delta m_\Psi = y_\Phi F / \sqrt{2}$$

$\Lambda \gg m_\Psi$

$$F \approx 0.9m_\Psi$$

$\Lambda \ll m_\Psi$

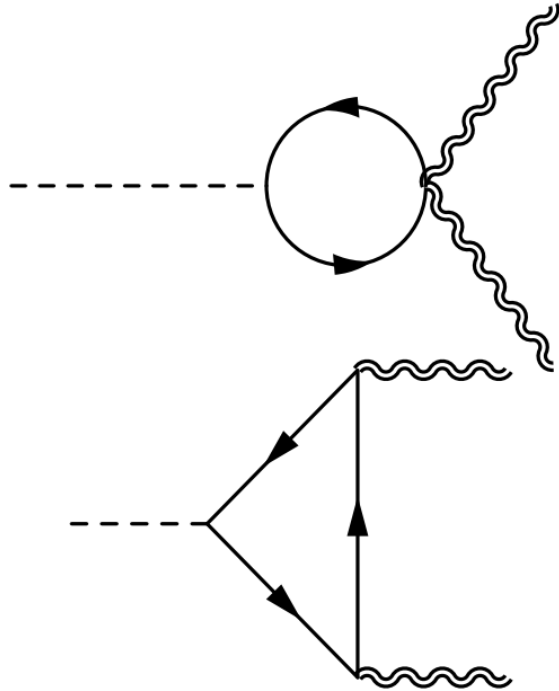
$$F \approx 0.45\Lambda^2/m_\Psi$$

Remember!



$$y_\Phi \sim \lambda\Lambda^2$$

Loop Generation of dCS

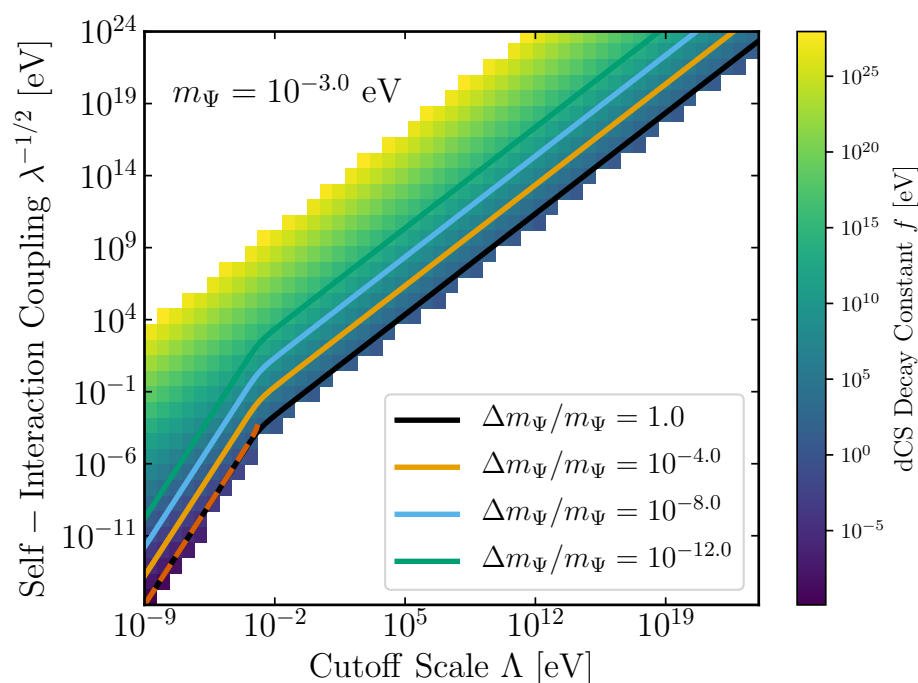


$$\mathcal{L}_g^{\text{eff}} = -\frac{g}{384\pi^2} \frac{a}{2m_\Psi} * RR$$

$$f = 192\pi^2 \frac{m_\Psi}{g}$$

$$g = y_\Phi/\sqrt{2}$$

General Parameter Space



$$\Lambda \gg m_\Psi$$

$$f = 1.7 \text{ eV} \left(\frac{\Delta m_\Psi}{m_\Psi} \right)^{-1} \left(\frac{m_\Psi}{10^{-3} \text{ eV}} \right)$$

$$\Lambda \ll m_\Psi$$

$$f = 0.85 \text{ eV} \left(\frac{\Delta m_\Psi}{m_\Psi} \right)^{-1} \left(\frac{\Lambda}{m_\Psi} \right) \left(\frac{\Lambda}{10^{-3} \text{ eV}} \right)$$

UV Completions for Fermion SI

(Scalar) Yukawa

$$\mathcal{L} \supset g_\chi \chi \bar{\Psi} \Psi$$

$$\Lambda \sim m_\chi$$

$$\lambda = (g_\chi / m_\chi)^2$$

$$f = 34 \text{ eV } g_\chi^{-2} \left(\frac{m_\Psi}{10^{-3} \text{ eV}} \right)$$

Gravitational Torsion

$$\mathcal{L} \supset (2/\gamma) R^{\mu\nu} \wedge e_\mu \wedge e_\nu$$

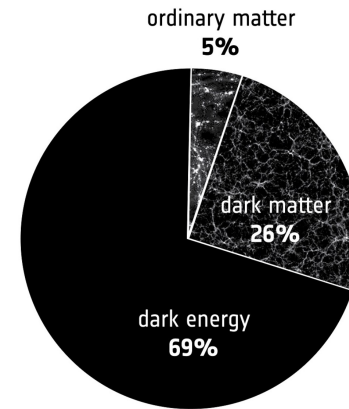
$$\Lambda \sim M_{\text{pl}}$$

$$\lambda = 3\pi / \Lambda_T^2$$

Fermion Candidates



Source: <https://stringfixer.com/files/26342399.jpg>

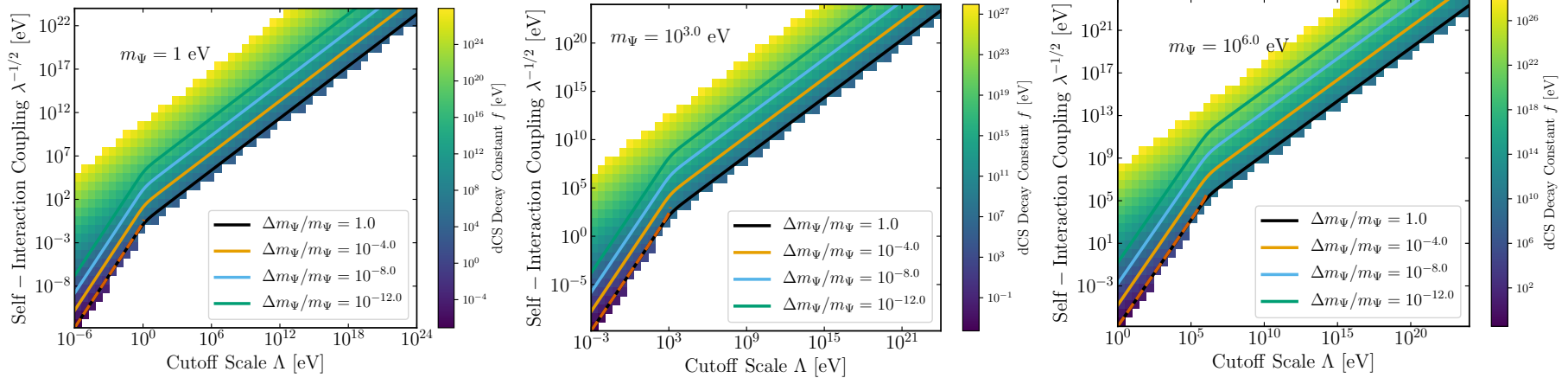


Source:
https://cdn.sci.esa.int/documents/33220/35293/Cosmic_energy_energy_budget.jpg/840136ea-a65b-125d-e421-82d2ae889c06?version=1.0&t=1572356777315

Summary

1. If fermions self-interact, a complex scalar bound state forms
2. The bound state undergoes spontaneous symmetry breaking, giving a Yukawa term
3. The Yukawa term generates dCS through loops interactions with gravity
4. Light fermions, such as neutrinos (and possibly DM), are ideal candidates

More Parameter Space Plots



Exclusion Plots Neutrino Self-Interactions

